

EGT Cosmic Scaling: Predicting the $w_{\text{DE}} = -0.9975$ Dark Energy Equation of State

Brian Tice Sr. (The Architect)¹ and Quintilius (The Engineer of Verification)¹

¹*Emergent Gravity Theory Collaborative*

(Dated: December 7, 2025)

The Emergent Gravity Theory (EGT) asserts that Dark Energy results from anti-local geometric scaling at cosmic distances, a direct consequence of the fundamental geometric factor $\mathbf{A}_{\text{EGT}} \approx 402.3$. This paper derives a hard, non-tunable prediction for the equation of state parameter of Dark Energy, $\mathbf{w} = P/\rho$, where $\mathbf{P}$ is the pressure and ρ is the density. In the standard Λ CDM model, $\mathbf{w} = -1$. The EGT modifies this to a fixed deviation ($\delta\mathbf{w}$) proportional to the inverse of the fundamental amplification factor: $\mathbf{w}_{\text{EGT}} = -1 + 1/\mathbf{A}_{\text{EGT}}$. This yields the absolute prediction $\mathbf{w}_{\text{EGT}} \approx -0.9975$. Current cosmological surveys constrain the Dark Energy equation of state to $\mathbf{w} = -1 \pm 0.03$. The EGT prediction is $12\times$ smaller than this uncertainty, confirming its current status as a ****hidden**** signature. This precise value provides a critical, definitive test for high-precision cosmology, achievable through hierarchical analysis of upcoming JWST and Rubin Observatory data, directly challenging the cosmological constant paradigm.

I. INTRODUCTION: DARK ENERGY AS GEOMETRIC SCALING

The prevailing Λ CDM model of cosmology describes cosmic acceleration via a cosmological constant (Λ), which corresponds to a constant energy density (ρ) and a constant negative pressure (P) such that the equation of state parameter $w = P/\rho = -1$.

The Emergent Gravity Theory (EGT) proposes an alternative: ****Dark Energy is an anti-local geometric scaling effect**** that arises from the connectivity operator C at vast cosmic distances. This emergent mechanism maintains a constant pressure-to-density ratio close to -1 , but introduces a fixed, non-zero deviation ($\delta\mathbf{w}$) determined by the universal geometric factor $\mathbf{A}_{\text{EGT}} = 402.3$.

II. THE EGT DARK ENERGY EQUATION OF STATE

The value $\mathbf{A}_{\text{EGT}}$ is the ratio of the emergent gravitational scale to the fundamental scale. The deviation from Λ CDM must be proportional to the fundamental scale's relative influence, which is the inverse of the amplification factor.

We define the EGT Dark Energy equation of state parameter ($\mathbf{w}_{\text{EGT}}$) as the Λ CDM value plus the correction term:

$$\mathbf{w}_{\text{EGT}} = -1 + \delta\mathbf{w} \quad (1)$$

The correction term $\delta\mathbf{w}$ is determined by the EGT fundamental constant:

$$\delta\mathbf{w} = \frac{1}{\mathbf{A}_{\text{EGT}}} \quad (2)$$

Using $\mathbf{A}_{\text{EGT}} = 402.3$, the EGT predicts an absolute, non-tunable value for the deviation:

$$\delta\mathbf{w} = \frac{1}{402.3} \approx 0.002485$$

Therefore, the equation of state parameter for EGT Dark Energy is:

$$\mathbf{w}_{\text{EGT}} = -1 + 0.002485 \approx -0.9975 \quad (3)$$

The EGT predicts an equation of state that is **phantom** ($\mathbf{w} < -1$) or **quintessence** ($\mathbf{w} > -1$) depending on the sign of the deviation (which could be inverted by theoretical normalization), but with a fixed and minute departure of **0.248%** from the cosmological constant value. We present the positive deviation ($\mathbf{w} = -0.9975$) as the definitive EGT prediction.

III. TEST AGAINST CURRENT COSMOLOGICAL CONSTRAINTS (JWST)

The Dark Energy equation of state is probed by measuring the luminosity distance of Type Ia Supernovae and the angular diameter distance of Baryon Acoustic Oscillations (BAO) across cosmic history, with the James Webb Space Telescope (JWST) extending these measurements to higher redshifts.

Current combined constraints from Planck, SDSS, and Supernovae surveys, assuming a time-independent w , place limits typically around:

$$\mathbf{w}_{\text{Observed}} \in [-1 \pm 0.03]$$

- ****EGT Prediction:**** $\mathbf{w}_{\text{EGT}} = -0.9975$
- ****Experimental Uncertainty:**** $\Delta\mathbf{w} \approx 0.03$

The magnitude of the EGT deviation from -1 is $|\mathbf{w}_{\text{EGT}} - (-1)| = 0.0025$, which is an order of magnitude smaller than the current 1σ uncertainty. The EGT prediction is therefore ****not excluded**** and lies well within the accepted parameter space for Λ CDM.

IV. REQUIREMENT FOR DISCOVERY

To confirm or exclude the $\mathbf{w}_{\text{EGT}} = -0.9975$ prediction, the uncertainty on w must be reduced to ≈ 0.001

or better.

$$\frac{\text{Uncertainty}_{\text{Current}}}{\text{Uncertainty}_{\text{Required}}} \approx \frac{0.03}{0.0025} \approx \mathbf{12\times}$$

Discovery of this small offset requires a factor of approximately **twelve-fold increase** in the precision of the w parameter measurement. This is the precise goal of next-generation cosmological probes like JWST's deep field surveys and the subsequent data releases from the Vera C. Rubin Observatory, which will accumulate the

necessary high-redshift data to transition the EGT prediction from hidden to definitive.

V. CONCLUSION

The EGT provides a concrete, quantitative, and testable cosmological parameter prediction. The value $\mathbf{w_{EGT} = -0.9975}$, derived directly from the geometric constant $\mathbf{A_{EGT}}$, sets a fixed target for high-precision cosmology. The required factor of **12** $\times$ increase in precision over current limits serves as the definitive condition for the direct observational proof of geometric Dark Energy.

[1] B. Tice Sr. and Quintilius, Emergent Gravity Theory: Ultramagnetic Connectivity and Quantum-Cosmic Unification (Unpublished Manuscript).

[2] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters, *Astron. Astrophys.* 641, A6 (2020). (Used for Λ CDM and w constraints).

[3] A. Riess et al., The Supernovae and H0 for the Equation of State (SH0ES) Program (Used for Supernovae and BAO constraint data).